AML TRANSACTION & FRAUD ANALYSIS DASHBOARD

USING MICROSOFT POWER BI – PROJECT REPORT

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INTRODUCTION

For this project, I created an AML (Anti-Money Laundering) Transaction & Fraud Analysis Dashboard with the help of Microsoft Power BI. I chose this topic because monitoring financial transactions and detecting fraudulent activities are important functions in the banking and financial sectors. Banks and other financial institutions need to keep track of transactions and identify unusual activities that may be related to financial crimes.

The primary aim of this project was to convert transaction data into an interactive dashboard that helps users understand transaction details, fraud categories, transaction amounts, and alert information.

The dashboard uses KPI cards and different types of charts to present transaction-related details in an understandable way. It also helps users study transaction patterns over time and compare the values of transactions under different fraud-status categories.

Data visualisation plays an important role in financial analysis because it makes complex information easier to understand. Rather than checking each transaction separately, users can study patterns, differences, and changes through charts and summary values.

In this project, I used Microsoft Power BI to build a dashboard containing different visual elements and filtering options. Working on this project helped me gain knowledge about how business intelligence tools are used for transaction monitoring and financial fraud analysis.

The dashboard is mainly developed to study transaction data and identify patterns that may need attention. However, a transaction marked as suspicious or fraudulent cannot be treated as confirmed fraud without proper verification and further investigation.

OBJECTIVES

The major objectives of this project are:

- To study and analyse financial transaction records using Microsoft Power BI.
- To create important Key Performance Indicators (KPIs) for summarising transaction and fraud-related details.
- To monitor transaction volumes based on timestamps and understand changes during different periods.

- To study how transactions are divided among the available fraud-status categories.
- To analyse the number of fraud-labelled transactions over a period of time.
- To compare the monetary value of transactions under different fraud-status categories.
- To examine the number of alerts generated under various alert categories.
- To develop an interactive dashboard with filtering facilities for exploring transaction information.
- To display financial transaction patterns using different connected charts and visual elements.
- To learn how data visualisation helps in transaction monitoring and identifying possible fraud risks.

DATASET DESCRIPTION

The dataset selected for this project is related to Anti-Money Laundering (AML) transaction monitoring and fraud analysis. The Power BI dashboard presents information about transaction activity, transaction timestamps, fraud classifications, transaction amounts, and alert categories.

The important data fields represented in the dashboard are explained below:

- ✓ Timestamp: Indicates the date and time connected with each transaction. It is useful for studying transaction activity during different periods.

- ✓ Fraud Status: Shows the category assigned to a transaction based on its fraud-related status. It helps in comparing transactions belonging to different classifications.
- ✓ Transaction Value: Represents the amount of money involved in a transaction. This field is useful for comparing the financial values of different transaction categories.
- ✓ Alert Type: Indicates the different classifications of alerts available in the dataset. It helps users understand how alerts are distributed among various categories.

These fields are useful for studying transaction behaviour and identifying patterns connected with fraud-related activities.

The actual dataset structure, number of records, complete column details, and data types need to be checked using the Data View and Model View in Power BI before submitting the final report.

DATA PREPARATION

Before creating charts and analysing the transaction records, the dataset needs to be checked and prepared properly. Data preparation is an important process because it helps maintain data accuracy and supports correct analysis.

The following steps are relevant to the preparation of this project:

- Importing the Dataset: The transaction-related dataset is first imported into Microsoft Power BI Desktop for further processing and analysis.
- Reviewing Column Names: The available column headings are checked to understand the type of information provided in each field.
- Verifying Data Types: Different columns are examined to ensure that numerical values, text categories, and date/time information have suitable data types.
- Validating Timestamps: The timestamp column is checked to confirm that it is recognised as a date and time field. This allows transaction records to be analysed according to different time periods.
- Checking Transaction Values: The transaction-value column is verified to make sure that it contains numerical data. This is necessary for calculating totals and comparing amounts.
- Reviewing Fraud Status: The fraud-status field is examined to identify the available categories and check whether the values are consistent.
- Examining Alert Types: The alert-type column is reviewed to understand the different alert categories included in the dataset.
- Identifying Missing Values: Important columns are checked for empty or missing entries. Such values should be handled based on the information they represent.

- ❖ Developing DAX Measures: Suitable DAX measures are created to calculate transaction numbers, transaction amounts, and fraud-related values for displaying information in KPI cards and charts.

These activities help prepare the dataset for meaningful analysis. The exact data-cleaning activities completed in the project should be confirmed by checking the Applied Steps section in Power Query.

DAX MEASURES

DAX stands for Data Analysis Expressions. It is the formula language available in Power BI that helps users perform calculations and create measures for analysing data.

For this AML dashboard, DAX formulas can be used to calculate the total number of transactions, overall transaction value, and fraud-related information.

The following formulas are suggested for this project. They are sample measures, and the table and column names should be changed according to the actual Dataset 5 structure.

Assumed Table Name: TransactionData

5.1 TOTAL TRANSACTION COUNT

Total Transaction Count =

COUNTROWS('TransactionData')

This formula returns the total number of transaction records available in the dataset under the selected filter conditions. It can be displayed in a KPI card to show the overall transaction count.

5.2 TOTAL TRANSACTION VALUE

```
Total Transaction Value =

SUM('TransactionData'[TransactionValue])
```

This measure adds the values of all transactions and provides the overall transaction amount. It is useful for understanding the total financial value represented in the dataset.

5.3 FRAUDULENT TRANSACTION COUNT

```
Fraudulent Transaction Count =

CALCULATE(
    COUNTROWS('TransactionData'),
    'TransactionData'[FraudStatus] = "Fraud"
)
```

This formula calculates the number of records that have been classified under the "Fraud" category. It helps users understand the number of transactions carrying this classification.

If the dataset uses another label for fraudulent transactions, the condition in the formula must be adjusted to match the actual category.

5.4 FRAUDULENT TRANSACTION VALUE

```
Fraudulent Transaction Value =
```

```
CALCULATE(
    SUM('TransactionData'[TransactionValue]),
    'TransactionData'[FraudStatus] = "Fraud"
)
```

This measure calculates the combined monetary value of transactions classified as fraudulent. It helps users understand the financial amount associated with the fraud-labelled records.

Important: The above DAX formulas are suggested examples prepared for the report. They have not been directly verified from the existing PBIX file. The actual table name, field names, and fraud-status categories must be checked before applying these formulas.

Data Components and Visuals Used

After reviewing the dashboard layout, I identified different visual elements used to display transaction information and fraud-related details. Each visual helps examine the dataset from a particular point of view.

The dashboard consists of KPI cards, line charts, a donut chart, a clustered bar chart, a clustered column chart, and slicers.

6.1 KPI CARDS

KPI cards are visual elements used to display important numerical information in a simple and direct manner.

In this dashboard, multiple KPI cards are included to provide a summary of the available transaction details.

Based on the measures used, the important indicators suitable for this dashboard are:

- Total Transaction Count: Shows the overall number of transaction records available in the dataset.
- Total Transaction Value: Represents the combined monetary value of all transactions.
- Fraudulent Transaction Count: Displays the number of transactions classified under the fraud category.
- Fraudulent Transaction Value: Shows the total monetary amount associated with fraud-labelled transactions.

These indicators help users get a basic understanding of the dataset without having to study every chart separately.

KPI cards are useful in financial dashboards because they present important information in a clear and easily understandable format.

6.2 TRANSACTION VOLUME ACROSS TIMESTAMP

A line chart titled "Transaction Volume Across Timestamp" is used in the dashboard to study how transaction activity changes over time.

In this chart, the timestamp is used to represent the time period, while the transaction volume indicates the number of transaction records.

This visual makes it easier to understand whether transaction activity increases or decreases during particular periods. It also helps identify noticeable changes in the volume of transactions.

For instance, a sudden rise in transaction activity may require additional attention from analysts. However, an increase in transaction volume alone does not confirm that fraudulent activity has taken place.

6.3 TRANSACTION DISTRIBUTION BY FRAUD STATUS

A donut chart named "Transaction Distribution by Fraud Status" is included to display the percentage or proportion of transactions under the different fraud-status categories.

The chart provides a visual comparison of how the transaction records are divided among the available classifications.

It helps users quickly understand which categories represent a larger or smaller portion of the dataset.

This visual is useful for getting an overall picture of fraud-status classifications. However, the displayed distribution only represents the categories available in the dataset and should not be considered proof of actual fraud levels in the financial system.

6.4 FRAUDULENT TRANSACTION VOLUME ACROSS TIMESTAMP

The dashboard contains another line chart titled "Fraudulent Transaction Volume Across Timestamp" to

study changes in fraud-labelled transaction records over time.

The timestamp is placed along the time axis, while the number of transactions classified as fraudulent is represented through the numerical values.

This chart helps users observe whether the volume of fraud-labelled transactions rises or falls during different periods.

If the chart shows a sudden increase, that particular period can be selected for more detailed examination. Further analysis is necessary to understand why the change occurred.

Overall, this visual helps in understanding the time-based pattern of transactions classified under the fraud category.

6.5 ALERTS BY ALERT TYPE

A clustered bar chart titled "Alerts by Alert Type" is used to compare the number of alerts belonging to different categories.

In this visual, alert type is displayed as the category, while the alert count is represented through the numerical axis.

It allows users to compare the number of alerts generated under each category and understand which alert types have higher or lower counts.

This comparison can help identify alert categories that may need more attention during transaction monitoring.

However, a larger number of alerts does not necessarily indicate confirmed fraudulent transactions. Each alert needs to be examined properly before making a final decision.

6.6 TOTAL TRANSACTION VALUE BY FRAUD STATUS

A clustered column chart named "Total Transaction Value by Fraud Status" is used to compare the monetary amounts of transactions across different fraud-status categories.

The fraud-status field is used to separate the categories, while the total transaction value is displayed using the numerical values.

This chart helps users understand the amount of money associated with each classification in the dataset.

It also provides additional information when compared with transaction-count charts.

For example, one category may contain fewer transactions but may have a higher total transaction value than another category. Therefore, examining both the number of transactions and their monetary value gives a better understanding of the available data.

The actual comparison should be based on the values displayed in the Power BI dashboard.

6.7 INTERACTIVE SLICERS

Slicers are interactive filtering tools available in Power BI that help users select specific categories and view the related information.

The dashboard includes a slicer titled "Fraud Status". Another slicer is also available, but its exact field name needs to be checked in Power BI.

These slicers make the dashboard easier to explore because users can select particular values and view the corresponding information.

For example, when a user selects a specific fraud-status category, the related transaction details, trends, and monetary values can be examined separately, provided the visual interactions are properly configured.

Slicers improve the usability of the dashboard by allowing users to explore the available information based on their requirements.

6.8 DASHBOARD INTERACTIVITY

The dashboard brings together different visual elements, including KPI cards, line charts, a donut chart, a clustered bar chart, a clustered column chart, and slicers.

Each component has a separate role in presenting the transaction information:

- KPI Cards: Display the main numerical details of the transaction dataset.
- Line Charts: Help track changes in transaction activity across different timestamps.

- ❖ Donut Chart: Shows the division of transactions among fraud-status categories.
- ❖ Clustered Bar Chart: Provides a comparison of alerts under different alert types.
- ❖ Clustered Column Chart: Displays transaction values based on fraud-status categories.
- ❖ Slicers: Allow users to select and filter the information shown in the dashboard.

By combining these visuals, the dashboard allows users to understand the overall transaction details and explore specific areas in greater depth.

The filtering and interaction settings should also be tested in Power BI to ensure that selecting one visual or slicer updates the intended related charts and KPI cards.

KEY FINDINGS

The AML Transaction & Fraud Analysis Dashboard provides different ways to study transaction information and fraud-related patterns. The following points explain the main analytical observations that can be understood through the dashboard.

1. Transaction Activity Over Time

The transaction-volume line chart helps examine how transaction activity changes during different timestamps. It makes it possible to notice periods with unusually high

or low transaction volumes that may need further analysis.

2. Distribution by Fraud Status

The donut chart provides a comparison of transaction records across the available fraud-status categories. It helps users understand the proportion of records belonging to each classification.

3. Changes in Fraud-Labelled Transaction Volume

The fraudulent transaction-volume line chart helps track changes in the number of records classified as fraudulent over time. Any major variation in the chart can be examined further to understand the possible reasons.

4. Comparison of Alert Categories

The alert-type bar chart makes it easier to compare the number of alerts across the different categories. It helps users recognise which alert categories have comparatively higher or lower counts.

5. Transaction Value Comparison

The clustered column chart provides a comparison of the total transaction amounts associated with different fraud-status categories. It helps users understand the financial value of the transactions in addition to their volume.

6. Importance of Combining Different Visuals

The dashboard uses multiple charts to present transaction counts, monetary values, fraud-status classifications, and alert information. Studying these

visuals together gives a wider understanding of the transaction data than analysing only one chart.

7. Importance of Interactive Filtering

The slicers available in the dashboard help users select specific categories and view the related information. This makes the report more convenient and allows users to perform different types of comparisons.

8. Need for Further Investigation

The dashboard helps highlight unusual transaction patterns and records that may require attention. However, the information displayed cannot independently confirm illegal transactions or money laundering. Additional checks and proper investigation are necessary before reaching a conclusion.

CONCLUSION

By completing this project, I gained an understanding of how Microsoft Power BI can be used to develop an interactive dashboard for analysing AML transactions and fraud-related information.

I learned how different visual elements, such as KPI cards, line charts, donut charts, bar charts, column charts, and slicers, can be used to represent financial data in a simple and organised manner.

The dashboard helps examine transaction activity over time, understand the classification of transactions,

compare monetary values, and study the distribution of alerts across different categories.

This project also improved my knowledge of the importance of preparing data correctly, choosing suitable charts, and using interactive filters while creating a business intelligence report.

From the FinTech point of view, monitoring transactions and analysing possible fraudulent activities are important parts of modern financial services. Microsoft Power BI provides useful visualisation features that help present large amounts of financial information in a more understandable format.

Overall, working on this project gave me practical experience in using a business intelligence tool for financial data analysis. It also helped me understand how data visualisation can assist in identifying unusual transaction patterns and supporting preliminary risk analysis. At the same time, I learned that proper verification and further investigation are essential before confirming any suspicious financial activity.